



INSTRUCTOR Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 20 Minutes

Quiz No 8

[20 points]

Note – Question No 2 is on page 2.

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [10 points] Cumulative Distribution Function for a Discrete Random Variables

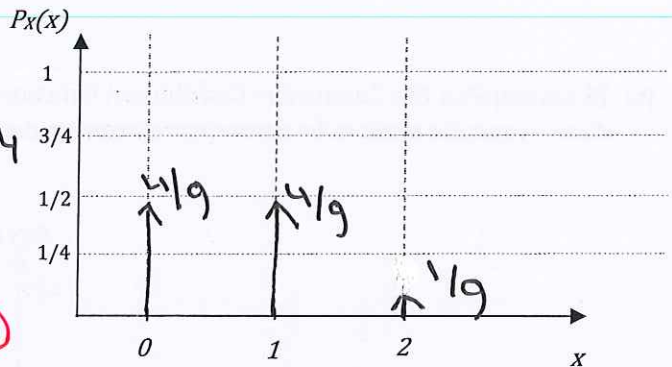
You toss a biased coin two times, noting the value of the DRV X (X is the number Heads in the two throws). Therefore, X is a Binomial Random Variable with parameter $n = 2$ with probability of success (Heads) = $1/3$.

(a) [3 points] Calculate and plot the P.M.F, $P_X(x)$ for the DRV X .

$$(a) P_X(0) = \binom{2}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^2 = \frac{4}{9} = 0.444 \quad (1)$$

$$P_X(1) = \binom{2}{1} \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9} = 0.444 \quad (1)$$

$$P_X(2) = \left(\frac{1}{3}\right)^2 = \frac{1}{9} = 0.11 \quad (1)$$



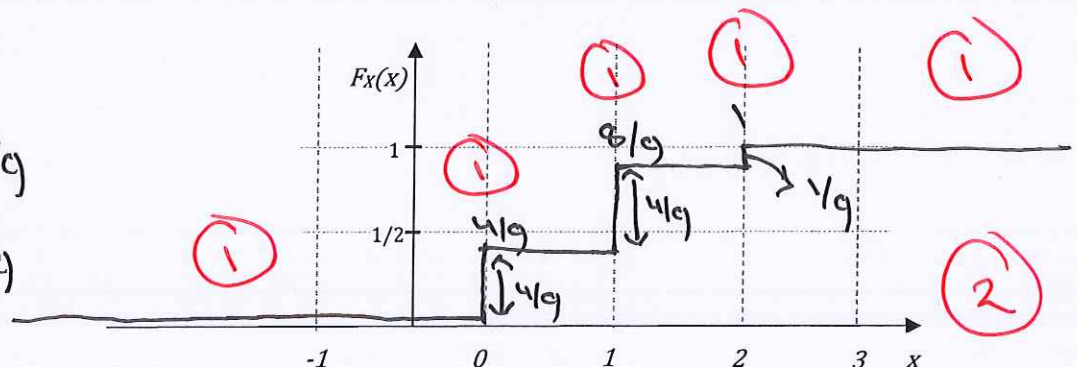
$$(b) F_X(-1) = 0$$

$$F_X(0) = 4/9$$

$$F_X(1) = 8/9$$

$$F_X(2) = 1$$

$$F_X(3) = 1$$



Question 2 [10 points] Cumulative Distribution Function for a Continuous Random Variables

Random Variable Y is a Standard Normal (Gaussian) Random Variable i.e. $Y \sim N(0, 1)$.

- (a) [5 points] Determine the Cumulative Distribution Function for the RV Y , $P(Y \leq y)$ for the following values of y . Write your answers in the given table.

$$\begin{aligned} P(Y \leq -3) &= 1 - P(Y \leq 3) \\ &= 1 - 0.9987 \\ &= 0.0013 \end{aligned}$$

$$\begin{aligned} P(Y \leq -1) &= 1 - P(Y \leq 1) = 1 - 0.8413 \\ &= 0.1587 \end{aligned}$$

S. No	y	CDF at $Y=y$, $P(Y \leq y)$
1	$-\infty$	0
2	-3	0.0013 (1)
3	-1	0.1587 (1)
4	0	0.5 (1)
5	+1	0.8413 (1)
6	+3	0.9987 (1)
7	$+\infty$	1

- (b) [5 points] Plot the Cumulative Distribution Function CDF, $P(Y \leq y)$ for the RV Y .

Note – your plot needs to be approximately correct, only.

